Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	24 October 2026
Team ID	
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.N.	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyse data.	Python, SQL HTML, CSS, etc.	Python handles the server routing; HTML/CSS provides the structure; JS handles the Tableau embed API.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	Snowflake / MySQL / Excel / CSV	Efficiently holds the athletes, games summary, and investment datasets locally.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Snowflake Stages / COPY INTO / Tableau Data Connections	Directly imports and relates CSV files using in-memory processing.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate	Eg: Tableau prep, Snowflake SQL, Tableau Desktop	Utilized for calculated fields, data relationship mapping, and

		KPIs, metrics, business rules, and analytical measures.		Parameter logic.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Snowflake SQL, Charts, Tableau , Python	Provides robust graphing Capabilities and a free hosting Environment for the final dashboards.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public, Flask, Streamlit	Creates a professional, responsive web wrapper To contextualize the data story.